0580_w20_qp_32_Q8

8 **(a)**

COMMONWEALTH

Lindon picks a letter at random from this word.

Explain why the probability that he picks a letter M is not $\frac{1}{10}$.

..... [1]

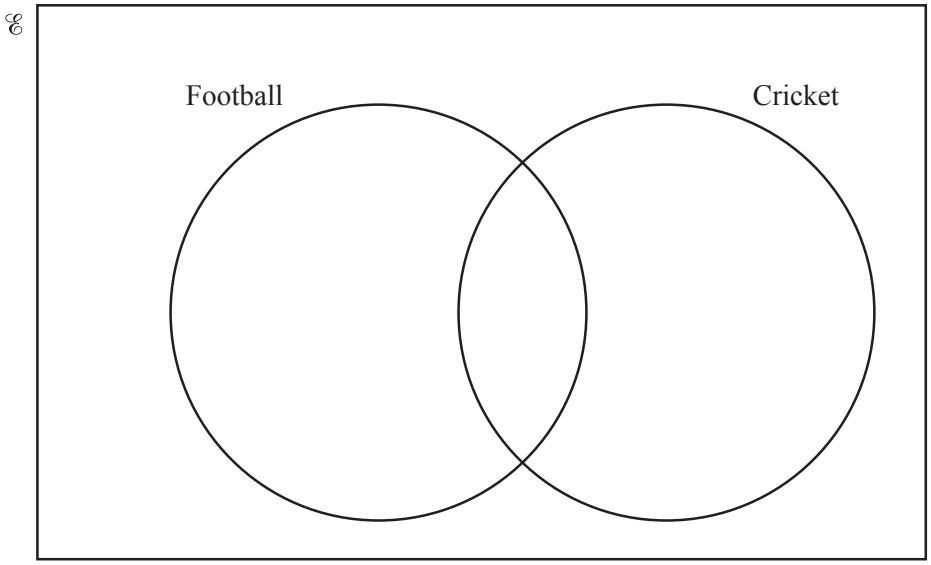
(b) Tickets for athletics or swimming or hockey or diving are placed in a box.
A ticket is picked at random from the box.

Sport	Athletics	Swimming	Hockey	Diving
Probability	0.12		0.09	0.4

Complete the table. [2]

(c) In a group of 40 students,

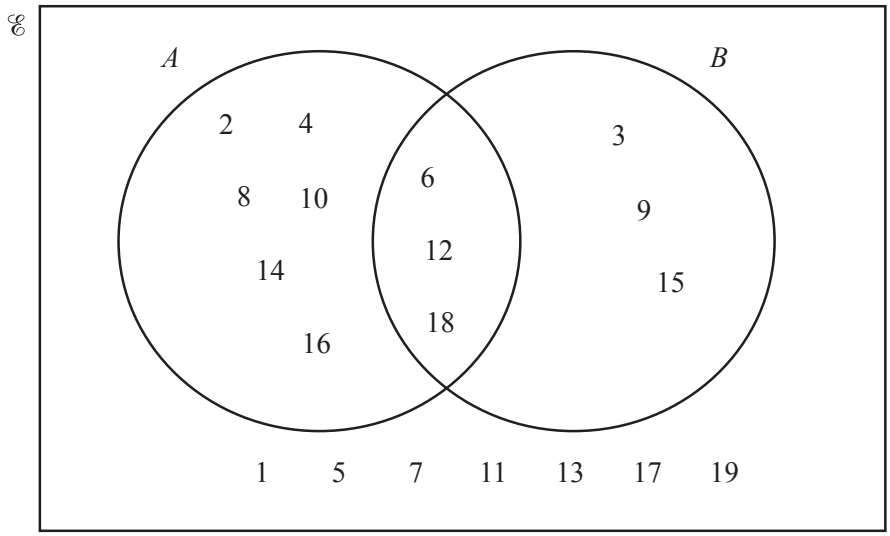
- 24 students like football
- 19 students like cricket
- 10 students like football but not cricket.



Complete the Venn diagram. [3]

0580_w20_qp_32_Q8

(d) $\mathscr{E} = \{x : x \text{ is a positive integer less than } 20\}$
$A = \{x : x \text{ is an even number}\}$
$B = \{x : x \text{ is a multiple of } 3\}$



(i) Write down $n(A)$.

..... [1]

(ii) List the elements of set B.

$B = \{$ ..... $\}$ [2]

(iii) One of these 19 numbers is picked at random.

Work out the probability that this number is

(a) not in set A and not in set B,

..... [1]

(b) in $A \cup B$.

..... [1]

(iv) Complete the statement.

$A \cap B = \{x : x \text{ is } \ldots\ldots\ldots\